

SMART TROLLEY

Project Report

Using ESP32, RFID, Firebase & BLE Beacons

Project Theme

IoT-based automated shopping, real-time billing, indoor aisle navigation, and QR-based payment

Prepared as a technical project report based on the supplied Smart Trolley User Manual

Department / Institution: _____

Team Members: _____

Academic Year: _____

1. Introduction

The Smart Trolley is an IoT-enabled shopping system designed to automate important parts of the retail shopping process. The supplied reference describes a system that combines RFID item identification, an ESP32 controller, a Firebase cloud database, BLE beacon navigation, an OLED display, push buttons, and QR-based payment. The purpose is to reduce manual billing and provide customers with aisle guidance using Bluetooth signal strength.

The ESP32 acts as the central controller. It coordinates product identification, display updates, sensor inputs, navigation information, and communication with the cloud database. Product information and billing information are maintained through Firebase, while BLE beacon RSSI is used to assist indoor aisle navigation.

2. Objectives

- Automate product identification during shopping using RFID.
- Retrieve product information from a Firebase cloud database.
- Update the customer's running bill in real time.
- Display product details and billing information on an OLED display.
- Provide indoor aisle navigation using BLE beacon signal strength (RSSI).
- Generate a QR code for payment after shopping.
- Update the shopping summary on the connected webpage after payment.
- Use sensors, indicators, and push buttons to support reliable trolley operation.

3. System Overview

Component	Function
ESP32	Central controller managing sensors, display, and cloud communication.
RFID Module (MFRC522)	Reads product RFID tags and identifies products.
IR Sensor	Detects item addition or removal.
OLED Display	Shows item details, total bill, navigation information, and QR code.
BLE Beacon	Transmits Bluetooth signals used for aisle navigation.
Firebase Database	Stores product details and billing/shopping information.
Push Buttons	Start/stop navigation and complete the shopping/payment process.
Buzzer & LEDs	Provide audible and visual status indications.

4. System Architecture

The system follows a controller-and-cloud architecture. The RFID module provides product identification to the ESP32. The ESP32 communicates with Firebase through Wi-Fi to obtain or update product and billing data. The OLED provides local feedback to the customer. BLE beacons positioned around aisles provide Bluetooth signals, and the ESP32 interprets their RSSI values for navigation assistance. Push buttons provide user commands for navigation and payment.

RFID Tags	→	RFID Reader	→	ESP32	→	Firebase
BLE Beacons	→	RSSI	→	ESP32	→	OLED
IR Sensor	→	ESP32	→	Buzzer / LEDs		
Payment Button	→	ESP32	→	QR Payment	→	Shopping Summary

5. Hardware Requirements and Setup

- ESP32 development board
- MFRC522 RFID reader and RFID tags
- OLED display using I2C (SDA and SCL)
- IR sensor for item addition/removal detection
- BLE beacon(s) placed at aisle entrances
- Push buttons for navigation and payment/checkout control
- Buzzer and LEDs for status indication
- 5V battery or adapter for power
- Jumper wires, breadboard/PCB, connectors, and mounting hardware

According to the supplied manual, the RFID module is connected through SPI, the OLED through I2C, and the IR sensor, buttons, buzzer, and LEDs are connected to GPIO pins. BLE beacons are positioned at aisle entrances. The trolley electronics are powered from a 5V battery or adapter.

6. Software Requirements

- Arduino IDE for ESP32 development and code upload.
- Wi-Fi support for cloud communication.
- Firebase configuration for product and shopping data.
- RFID library for MFRC522 communication.
- OLED display library for displaying user information.
- BLE functionality for beacon scanning and RSSI-based navigation.

The reference manual lists WiFi.h, FirebaseESP32.h, Adafruit_SSD1306.h, MFRC522.h, and BLEDevice.h as required libraries. The exact library versions and credentials are project-configuration details and should be set according to the actual ESP32 software environment.

7. Working Principle

1. Each product is associated with an RFID tag and corresponding product details in Firebase.
2. When the RFID tag is scanned, the ESP32 identifies the product and retrieves its information.
3. The IR sensor is used to confirm item addition or removal.
4. The OLED display shows the item name, price, and updated total bill.
5. BLE beacon navigation uses received signal strength (RSSI) to assist the customer in locating aisles.
6. When shopping is complete, the Payment button is pressed to generate a QR code.

7. The customer completes payment using a UPI application.
8. The shopping/payment summary is updated on the webpage through Firebase.

8. BLE Navigation Operation

BLE beacons are placed at aisle entrances. The trolley scans for beacon signals and evaluates RSSI, or received signal strength indicator. As the trolley approaches a beacon, the signal strength changes. The ESP32 can therefore use RSSI as an indication of relative proximity and present navigation information to the customer through the OLED display.

The supplied manual specifically states that BLE beacon navigation helps locate aisles using RSSI strength. It does not specify a complete multi-aisle routing algorithm; therefore, the exact RSSI thresholds and route-selection logic are implementation-specific.

9. Firebase Cloud Database

Firebase provides the cloud-data layer of the Smart Trolley. Product information can be associated with RFID identifiers, while billing and shopping information can be stored for later viewing through the webpage. The ESP32 communicates with Firebase through Wi-Fi, allowing product and transaction-related information to be synchronized during operation.

Typical data flow:

RFID scan → ESP32 identifies tag → Firebase product lookup → product name/price returned → bill updated → OLED display updated → payment completed → shopping summary updated.

10. Payment Process

1. The customer completes product selection and verifies the displayed bill.
2. The Payment/checkout button is pressed.
3. The system generates a QR code for payment.
4. The customer scans the QR code using a UPI application.
5. After payment, the shopping summary is updated on the webpage through Firebase.

11. Operating Procedure

1. Power on the Smart Trolley.
2. Wait for the ESP32 and display system to initialize.
3. Scan products using the RFID reader.
4. Check the OLED display for item and total-bill updates.
5. Use the Navigation button when aisle guidance is required.
6. Continue shopping and verify the running total.
7. Press the Payment/checkout button when shopping is complete.
8. Scan the generated QR code with a UPI application and complete payment.
9. Confirm the shopping summary on the webpage.

12. Troubleshooting

Issue	Possible Cause	Suggested Solution
OLED not displaying	Loose I2C connection	Check SDA/SCL wiring and power.
RFID not scanning	Tag too far away or damaged	Bring the tag closer or replace the tag.
BLE navigation inaccurate	Weak beacon signal / placement	Adjust beacon placement and test RSSI.
Firebase not updating	Wi-Fi disconnected	Reconnect to Wi-Fi and retry.

13. Safety Precautions

- Avoid exposing electronic sensors and modules to moisture.
- Handle the ESP32, RFID reader, OLED, and other electronic components carefully.
- Ensure a stable power supply to prevent unexpected resets.
- Check wiring before applying power.
- Secure modules and wires properly when mounting them on a trolley prototype.

14. Key Features and Benefits

- Automated RFID-based product identification.
- Real-time bill calculation and display.
- Cloud-based product and shopping information through Firebase.
- Indoor aisle assistance using BLE beacon RSSI.
- QR/UPI-based payment workflow.
- Local visual feedback through an OLED display.
- Sensor, button, buzzer, and LED support for user interaction.

15. Scope and Limitations

The supplied reference establishes RFID scanning, IR-based item confirmation, BLE beacon navigation, Firebase communication, OLED feedback, and QR-based payment as the core functions. It does not provide detailed mechanical construction, exact electrical pin mapping, a complete database schema, exact BLE threshold values, or a guaranteed payment-verification mechanism. These details must be finalized during the actual hardware/software implementation and testing.

16. Conclusion

The Smart Trolley demonstrates how embedded systems and cloud technologies can be combined to improve the shopping workflow. The ESP32 coordinates RFID identification, sensor inputs, OLED feedback, BLE navigation, and Firebase communication. Product identification and billing can be handled electronically, while BLE beacons provide aisle navigation and QR-based payment supports a digital checkout process. Overall, the project serves as an IoT-oriented prototype for automated and connected retail shopping.

17. Reference Used

Primary source: supplied document, "User Manual – Smart Trolley Using ESP32, RFID, Firebase & BLE Beacons," 3 pages. This report preserves the source document's system terminology and expands it into a

project-report structure without adding unsupported implementation specifications.